

U24CS651- PROMPT ENGINEERING

HEALTH CARE

PROMPT

Role: You are a senior community nurse conducting a health awareness seminar for local families.

Task: Explain "**The Importance of Preventive Healthcare and Regular Check-ups**" in a clear and encouraging way.

Context: This is for a general audience of adults who may be hesitant to visit the doctor unless they feel sick.

Instructions:

1. Introduce Preventive Healthcare.
2. Explain what "Preventive Care" actually means.
3. Describe why it is better than "Curative Care" (treating after sickness).
4. Explain the types of preventive measures (Screenings, Vaccinations, Lifestyle).
5. Explain the role of "Early Detection" in serious illnesses.
6. Give real-life examples of routine check-ups.
7. Explain how preventive care saves time, money, and stress.
8. Mention common barriers (fear, cost) and how to overcome them.
9. End with a short summary.

Constraints:

1. Use warm, simple English.
2. Keep a professional yet caring tone.
3. Avoid scary medical statistics.

Output format:

1. **Title:** One clear, positive line
2. **Introduction:** Why preventive healthcare and regular check-ups matter
3. **Body:** Key points with examples (preventive care, prevention vs cure, measures, early detection, benefits, barriers)
4. **Conclusion:** Short, hopeful call to action
5. **Word Limit:** 200–300 words

Output:

 **The Power of Prevention: Caring for Your Health Before Problems Begin**

Introduction

As a community nurse, I often meet people who only visit the doctor when they feel unwell. While that's understandable, preventive healthcare helps us stay healthy and avoid serious

problems before they start. Regular check-ups are not just for when we are sick — they are an important way to protect our long-term health.

Body

What is Preventive Care?

Preventive care means taking steps to stay healthy and catch problems early. It includes regular check-ups, health screenings, vaccinations, and healthy lifestyle choices.

Prevention vs. Cure

Curative care treats illness after it appears. Preventive care works earlier — stopping illness before it becomes serious. It is often easier, safer, and more comfortable to prevent a disease than to treat it later.

Types of Preventive Measures

- **Screenings:** Blood pressure checks, blood sugar tests, cholesterol checks, mammograms, and colon screenings.
- **Vaccinations:** Flu shots and other recommended vaccines to protect against infections.
- **Healthy Lifestyle:** Balanced diet, exercise, good sleep, and stress management.

The Importance of Early Detection

Early detection can make a big difference in conditions like cancer, diabetes, and heart disease. When found early, treatment is often simpler and more successful.

Real-Life Examples

Annual physical exams, dental cleanings, eye tests, and routine women's or men's health exams are all examples of preventive care.

Saving Time, Money, and Stress

Preventive care reduces emergency visits, expensive treatments, and long recovery times.

Overcoming Barriers

Fear and cost can stop people from seeking care. Talk openly with your provider about concerns, and ask about community clinics or insurance coverage options.

Conclusion

Your health is your greatest asset. Small, regular check-ups today can protect you for years to come. Take that step — your future self will thank you.

EDUCATION

PROMPT

Role: Imagine as you are a teacher guiding a students to learn react Course in 30 days.

Task: Design a complete 30 days learning roadmap for React course with 2hrs study per day.

Context: The roadmap is fully designed for beginners with no prior experience in React and It's focus on covering all the basic concepts to advanced topics

Instructions:

- 1) Provide a roadmap for 30 days with 2hrs of study per day
- 2) Present the daily learning content in bullet points, with each topic explained in simple language.
- 3) Avoid technical jargon and keep explanations clear and beginner-friendly.
- 4) Explain the each topic with the relevant examples
- 5) everyday can solve the two or more task
- 6) weekend to create a mini project or practice a task to based on previous learned
- 7) Progress gradually from basic concepts to advanced topics.
- 8) Provide a final project at the end of the course to apply all skills.

Tone: formal and professional tone

Output

30-Day React Learning Roadmap (2 Hours Per Day)

Week 1: JavaScript & React Basics

Day 1: Introduction to React & Setup

- What is React?
 - A JavaScript library for building user interfaces (web apps).
 - Example: Creating a simple website with reusable parts like header and footer.
- Install Node.js and create a React app.
- Understand project folder structure.

Tasks:

- Install Node.js and VS Code.
- Create your first React app.
- Run the app and explore folders.

Day 2: Understanding Components

- Components are reusable building blocks.
- Example: A “Header” component and a “Footer” component.

Tasks:

- Create Header and Footer components.
- Display them in the main App.

Day 3: JSX (Writing HTML in React)

- JSX allows writing HTML-like code in JavaScript.
- Example: `<h1>Hello Student</h1>`

Tasks:

- Create a page with headings and paragraphs.
- Add an image using JSX.

Day 4: Props (Passing Data)

- Props pass data from parent to child.
- Example: Greeting message with different names.

Tasks:

- Create a Greeting component with name prop.
- Display 3 different greetings.

Day 5: State (Storing Data)

- State stores changing data.
- Example: Counter increasing number.

Tasks:

- Build a counter app.
- Add increase and decrease buttons.

Day 6 (Weekend Mini Project): Simple Profile Card

- Create profile card with name, image, and description.
- Use components, props, and simple styling.

Day 7 (Weekend Practice): Review & Practice

- Practice components, props, state.
- Modify profile card with extra details.

Week 2: Events, Forms & Lists

Day 8: Event Handling

- Respond to user actions (clicks).
- Example: Show alert on button click.

Tasks:

- Create button click event.
- Create show/hide text feature.

Day 9: Forms in React

- Handling input fields.
- Example: Simple login form.

Tasks:

- Create input field for name.
- Show entered text below form.

Day 10: Controlled Components

- Input value controlled by state.

Tasks:

- Create form with two inputs.
- Reset form button.

Day 11: Lists & Keys

- Display multiple items.
- Example: To-do list items.

Tasks:

- Display array of fruits.
- Add unique key for each item.

Day 12: Conditional Rendering

- Show content based on condition.
- Example: Login/Logout button.

Tasks:

- Toggle login state.
- Display message conditionally.

Day 13 (Weekend Mini Project): To-Do List App

- Add tasks.
- Delete tasks.
- Mark as completed.

Day 14: Practice & Improve To-Do App

- Add edit feature.
- Improve UI design.

Week 3: Routing & Advanced Concepts

Day 15: Introduction to Routing

- Navigate between pages.

- Example: Home, About, Contact pages.

Tasks:

- Install React Router.
- Create 3 pages.

Day 16: Navigation Links

- Add menu navigation.
- Switch between pages without reload.

Tasks:

- Add navigation bar.
- Highlight active page.

Day 17: useEffect (Side Effects)

- Run code after page loads.
- Example: Fetch data from API.

Tasks:

- Log message on load.
- Fetch sample API data.

Day 18: Fetching Data

- Get data from free API.
- Example: Display user list.

Tasks:

- Fetch data from API.
- Show loading message.

Day 19: Reusable Components

- Create reusable button.
- Example: Custom Button component.

Tasks:

- Build reusable card component.
- Use it multiple times.

Day 20 (Weekend Mini Project): User Directory App

- Fetch users from API.
- Display in cards.
- Add search feature.

Day 21: Practice & Debugging

- Fix errors.
- Improve project structure.

Week 4: Advanced Topics & Final Project**Day 22: React Hooks Overview**

- useState & useEffect revision.
- Introduction to useContext.

Tasks:

- Share theme using context.
- Toggle light/dark mode.

Day 23: Context API

- Avoid passing props many times.
- Example: User login info shared globally.

Tasks:

- Create context for user.
- Display user name in multiple components.

Day 24: Performance Basics

- Avoid unnecessary re-render.
- Example: React.memo concept (simple explanation).

Tasks:

- Wrap component using memo.
- Observe behavior.

Day 25: Custom Hooks

- Reusable logic.
- Example: useFetch hook.

Tasks:

- Create custom hook for fetching data.
- Use in two components.

Day 26: Project Planning Day

- Plan final project structure.
- Create layout design.
- Divide into components.

Day 27–29: Final Project Development

Final Project: Complete Task Management App

Features:

- User login (simple).
- Add, edit, delete tasks.
- Filter tasks.
- Dark/light mode.
- Routing between pages.
- Fetch sample data.

Daily Goals:

- Day 27: Setup & layout.

- Day 28: Add core features.
- Day 29: Add advanced features & polish UI.

Day 30: Final Review & Deployment

- Test application.
- Fix bugs.
- Deploy using free platform (e.g., Vercel or Netlify).

Tasks:

- Deploy project.
- Share project link.
- Review entire 30-day journey.

Conclusion

By following this 30-day roadmap with consistent 2-hour daily study, you will progress from beginner to confidently building full React applications. Practice daily, build projects regularly, and focus on understanding concepts through examples. Consistency is the key to mastering React successfully.